

Integrating Digital Product Passports with digital twins and IoT for lifecycle traceability of batteries and electronics to meet EU ESPR compliance.

Manufacturing × Product traceability and digital product passport × Digital twin · OS019 v3 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
60 is the world moving	36 can we play, can we win	MEDIUM 1 named in the evidence	€277m partial evidence · per year	NEXT regulation adopted or propose...	L3 17 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

The opportunity

This opportunity helps manufacturers of batteries and electronics meet the EU ESPR mandate for Digital Product Passports by integrating them with digital twins and IoT for dynamic lifecycle traceability. It targets compliance officers and sustainability leads in manufacturing, who must act now because the regulation is adopted and technical standards are in place, creating an urgent need for machine-readable product lifecycle data.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	€361m €45.8m – €2.4bn	€277m €35.2m – €1.8bn	€346k €44k – €2.3m	partial
Observed: tendered contract value, annualised	€20.9m €20.9m – €56.6m	€0	€0	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Manufacturing (DE)	51,224 enterprises	observed	Eurostat · sbs_sc_oww	2024
Enterprises adopting Digital twin	10.4 % of enterprises (10+ employees)	proxy	Eurostat · isoc_eb_iotn2 · E_IOTDPP	2021
Annual value of one engagement	€337k per enterprise per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-05
Size-weighted buyer base	10,301 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	0.1 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Adoption rate is a declared proxy. Internet of Things by NACE Rev. 2 activity series E_IOTDPP is used as a proxy for Digital twin: IoT in production as the observable precursor. Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders.

Observed: tendered contract value, annualised

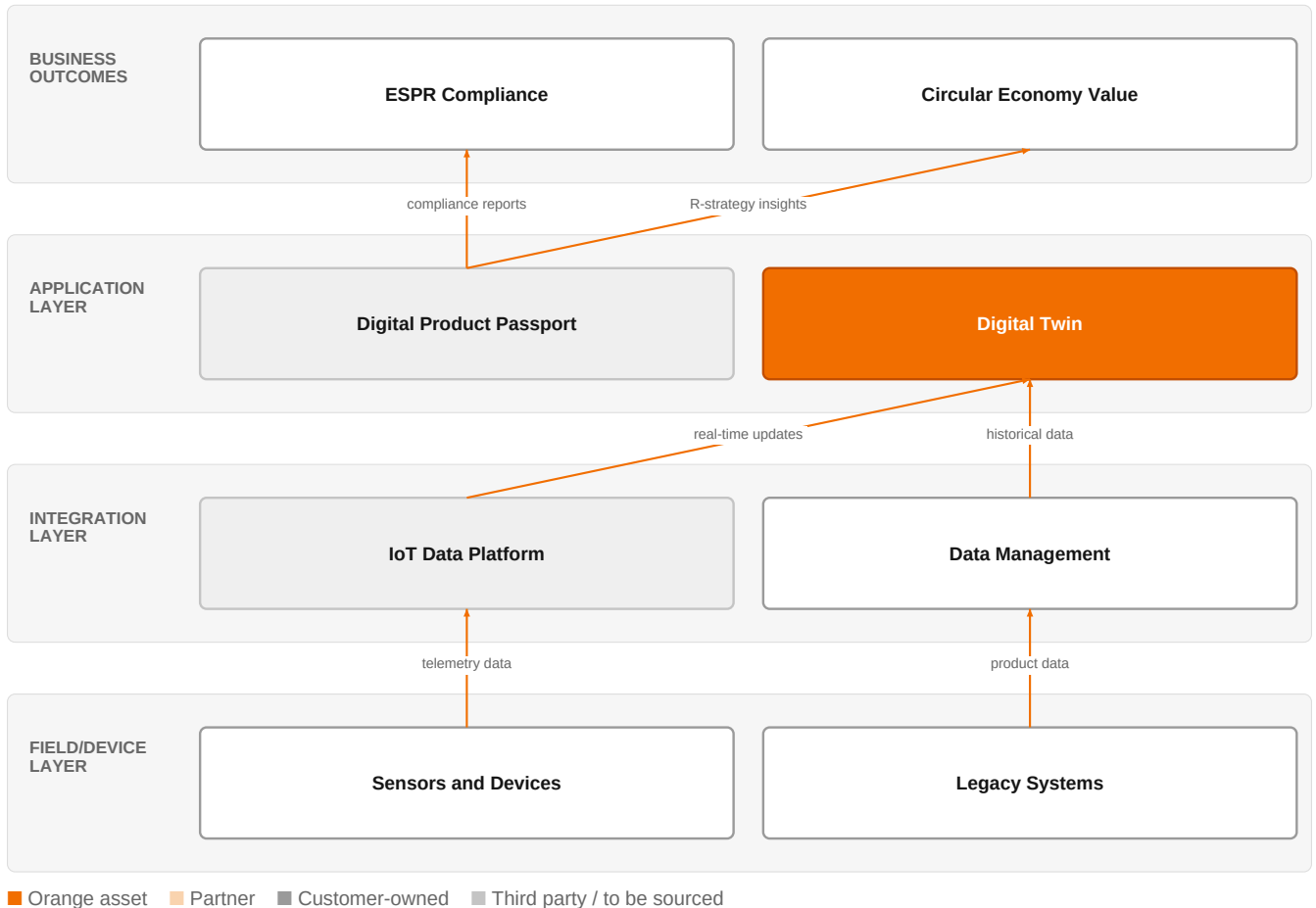
Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€20.9m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-05
Assumed obtainable share of the serviceable market	0.1 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 14 matching notices, matched on vertical via the CPV crosswalk — §4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a

market size.

The solution, in one picture

DPP Integration with Digital Twin and IoT



This diagram shows how Orange integrates IoT and digital twin assets to feed a compliant Digital Product Passport that drives both regulatory compliance and circular economy value.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

The EU's Ecodesign for Sustainable Products Regulation (ESPR) mandates Digital Product Passports for electronics and appliances by 2027, creating an urgent need for machine-readable product lifecycle data. European technical standards for DPPs are now in place, and the EU DPP registry has gone live, signaling that the infrastructure is maturing. However, effective DPP implementation requires robust integration of technologies to overcome scalability, data integrity, and legacy system management challenges, and the automotive sector specifically faces fragmented standards and slow adoption of digital tools.

Sources: SIG-BA196FF05244 Springer Science and B 2026-02-03; SIG-AF45FABC720F RECYCLING magazin 2026-08-03; SIG-370D30A66C8C EU NEIGHBOURS east 2026-07-21; SIG-F47C31AB60CB MDPI AG 2026-02-14; SIG-FDD8BAC05BAA cordis.europa.eu 2026-07-01

Who buys, and why now

The chief sustainability officer or head of compliance typically signs, while the operations and IT teams feel the pain of manually compiling lifecycle data across fragmented supply chains. The trigger is the ESPR compliance deadline, which forces them to produce DPPs for their products, but they lack the systems to collect and share data dynamically. They need to avoid the operational burden of non-compliance and the risk of market access restrictions.

Sources: SIG-BA196FF05244 Springer Science and B 2026-02-03

Why it is hot now — every claim bound to a dated source

- Digital Product Passports can represent executable circular economy R-strategies using behavior trees to support traceability in automation systems. [SIG-D0815CA3A107]

- Circular Economy R-strategies such as repair, reuse, remanufacturing, and recycling are critical for extending product lifecycles and reducing material waste. [SIG-D0815CA3A107]
- Behavior trees embedded in DPPs enable executable circular economy R-strategies for repair, reuse, remanufacturing, and recycling. [SIG-D0815CA3A107]
- Behavior Trees embedded in DPPs support traceability and automation of circular economy R-strategies. [SIG-D0815CA3A107]
- DPPs enable transparency and data sharing across a product's life cycle, facilitating circular strategies. [SIG-CC28A63D0BB7]
- Digitalization is transforming maintenance into autonomous, data-driven systems, improving performance and reducing costs. [SIG-41FB7CD0130C]
- The Digital Product Passport requires robust integration of technologies to overcome scalability, data integrity, and legacy system management. [SIG-F47C31AB60CB]
- Dynamic traceability and reliability management integrates Digital Product Passport with Digital Twins and IoT for batteries and electronics, addressing circular economy mandates. [SIG-F47C31AB60CB]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble a solution using its IoT experts and Orange Group researchers to design and deploy a DPP system integrated with digital twin technology. The engagement would start with a discovery phase to map the customer's product lifecycle data sources, then build a digital twin layer that ingests IoT data for dynamic traceability, and finally implement the DPP registry connection. Orange's ISO 27001 and TISAX certifications would underpin the security and automotive-specific compliance, and references like Colgate-Palmolive and Saint-Gobain Glass demonstrate cross-industry experience.

Why Orange can win

Orange can win because it combines IoT expertise with research capabilities and holds ISO 27001 and TISAX certifications, which are critical for manufacturing customers. Its digital twin technology is a named asset, and its references show it can deliver complex industrial solutions. However, the asset list is thin on specific DPP or circular economy references, so Orange must leverage its broader industrial track record and security credentials to differentiate.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Digital twin	technology	L3	no offer path, but adjacent assets or Orange research exist	unconfirmed
ISO 27001	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
TISAX	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Colgate-Palmolive	reference	SUP	published reference in this vertical, different use case	unconfirmed
Mondelez International	reference	SUP	published reference in this vertical, different use case	unconfirmed
Saint-Gobain Glass	reference	SUP	published reference in this vertical for this use case	unconfirmed
Skytale	reference	SUP	published reference in this vertical, different use case	unconfirmed

6 further published reference(s) in this vertical are linked to this topic and are listed in the radar.

Competitive landscape

MEDIUM competitive intensity (score 6.2; 1 of 8 listed players appear in this topic's own evidence)

A real field, including at least one player the customer will already know. Expect to be compared.

Competitors include Databricks, which is named in the evidence for its data and AI platform, and global systems integrators like Accenture and Capgemini that sell digital twin solutions. Industrial specialists such as Siemens, Schneider Electric, and PTC also compete with digital twin offerings, while Dassault Systemes/3DS Outscale provides sovereign cloud options. Deutsche Telekom/T-Systems is a telecom peer active in manufacturing. Customers will compare Orange's ability to integrate IoT and digital twin with the depth of these specialists.

Sources: SIG-BC952EFA76FC Databricks 2026-07-27

Competitor	Category	Position here	Basis	Evidence
Databricks	Data and AI platform	named in this topic's evidence	evidenced	SIG-BC952EFA76FC
Accenture	Global systems integrator	sells Digital twin; active in Manufacturing; competes in OX: Smart Industries	structural	—
Capgemini	Global systems integrator	sells Digital twin; active in Manufacturing; competes in OX: Smart Industries	structural	—
Dassault Systemes / 3DS Outscale	Sovereign cloud provider	sells Digital twin; active in Manufacturing; competes in OX: Smart Industries	structural	—
PTC (ThingWorx)	Industrial / OT specialist	sells Digital twin; competes in OX: Smart Industries	structural	—
Schneider Electric	Industrial / OT specialist	sells Digital twin; active in Manufacturing; competes in OX: Smart Industries	structural	—
Siemens	Industrial / OT specialist	sells Digital twin; active in Manufacturing; competes in OX: Smart Industries	structural	—
Deutsche Telekom / T-Systems	Telecom operator peer	active in Manufacturing; competes in OX: Smart Industries	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 What is your current approach to collecting and sharing product lifecycle data for your batteries or electronics?
- 2 Have you started any pilot or project to prepare for the ESPR Digital Product Passport mandate?
- 3 Which systems do you currently use for product data management, and how do they integrate with your IoT or sensor infrastructure?
- 4 Who in your organization is responsible for ESPR compliance, and what is their timeline for having DPPs in place?
- 5 Do you have a digital twin initiative already, and if so, how is it being used for product traceability?
- 6 What are your biggest pain points in achieving lifecycle traceability across your supply chain?

Objections you will meet

They will say	You can answer
We already have a digital twin platform from a major vendor; why would we need Orange?	That's a fair point, and you may have a strong platform. But our value is in integrating that platform with IoT data and DPP requirements, leveraging our IoT experts and research capabilities to ensure dynamic traceability and compliance. We can complement your existing investment rather than replace it.
The ESPR regulation is still evolving; we can wait until the standards are final.	The regulation is adopted, and technical standards are already in place. The registry is live, and early movers will have a competitive advantage in shaping their data models. Waiting could put you behind on compliance and miss the opportunity to use DPPs for circular economy value beyond compliance.
We have limited budget for sustainability projects; this seems like a large investment.	We understand budget constraints. Our approach is modular, starting with a discovery phase to identify quick wins and then scaling. We can also highlight how DPPs can enable cost savings through better end-of-life strategies, such as repair and recycling, which can offset the initial investment.

Risks and unknowns

The main risk is that the customer may already have a digital twin platform from a competitor, making Orange's integration role secondary. Another risk is that the DPP standards are still evolving, and the customer may delay investment until the final technical specifications are fully settled. What is not yet known is how the EU DPP registry will enforce data quality and whether the market will accept third-party DPP solutions over in-house builds.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Prepare a bid brief for a European automotive manufacturer, differentiating Orange's approach by combining Digital Product Passports with digital twins and IoT, and referencing our work with Saint-Gobain Glass and Colgate-Palmolive to demonstrate proven capability in lifecycle traceability.
Sales	In a meeting with a European automotive manufacturer's head of sustainability, say: 'Orange can help you meet EU ESPR compliance by integrating Digital Product Passports with digital twins and IoT for lifecycle traceability, leveraging our IoT expertise and ISO 27001 and TISAX certifications to ensure data integrity and security.'
Strategist / Innovator	Deep-dive on how Orange's IoT experts and Group researchers can integrate Digital Product Passports with digital twins and IoT for lifecycle traceability, focusing on scalability, data integrity, and legacy system management, and identify a pilot opportunity in the European automotive sector.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-FDD8BAC05BAA	2026-07-01	cordis.europa.eu	1	Driving the Circular Transformation of the European Automotive Industry with Skills, Digital To...
SIG-36FFB2435F8A	2026-04-20	openalex.org	1	A Comprehensive Review of Digital Product Passports in Sustainable Manufacturing: Current State...
SIG-F47C31AB60CB	2026-02-14	MDPI AG	1	Dynamic Traceability and Reliability Management: Integrating the Digital Product Passport with ...
SIG-BA196FF05244	2026-02-03	Springer Science and Busi...	1	Digital Product Passport for Circular Economy: A Review of Semantic Web Vocabularies and Requir...
SIG-4FE21FD2ED06	2026-01-01	openalex.org	1	On the Utilization of Digital Product Passports for Intelligent Metal Sorting and Recycling
SIG-CC28A63D0BB7	2026-01-01	openalex.org	1	From Compliance to Circularity: Stakeholder Perspectives on Adopting Digital Product Passports ...
SIG-D0815CA3A107	2026-01-01	openalex.org	1	Representing Executable Circular Economy R-Strategies using Behavior Trees Embedded in Digital ...
SIG-DFE69DA138F9	2026-01-01	openalex.org	1	Shaping a circular future for electronics: The role of Digital Product Passports
SIG-8123897AE6E6	2025-12-15	openalex.org	1	Bridging digital tools and human skills: A sustainable competence framework for implementing di...
SIG-6786E4AD1713	2025-11-22	openalex.org	1	The quest for more circular battery value chains: Implementing the EU digital battery passport ...
SIG-41FB7CD0130C	2025-10-21	openalex.org	1	Advanced MRO Processes in Industry 4.0 with Proactive Asset Administration Shell and Digital Pr...
SIG-EC1CEFFD337A	2025-10-15	openalex.org	1	Digital Product Passport in support of Data-Driven End-of-Use Strategies—a systems design persp...
SIG-40DC35D6534F	2025-10-14	openalex.org	1	Predictive Generation of Digital Product Passport as Decision Support in Circular Networks
SIG-6FBECA0F7EA7	2025-09-10	openalex.org	1	Design of a Sensor-Based Digital Product Passport for Low-Tech Manufacturing: Traceability and ...
SIG-AF45FABC720F	2026-08-03	RECYCLING magazin	2	Digital Product Passport: European Technical Standards Now in Place - RECYCLING magazin
SIG-BC952EFA76FC	2026-07-27	Databricks	2	The EU Digital Product Passport: a traceability deadline - Databricks
SIG-370D30A66C8C	2026-07-21	EU NEIGHBOURS east	2	EU's Digital Product Passport Registry goes live - EU NEIGHBOURS east

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: diagram (2 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:15+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:18+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-18T20:41:29+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.